

Name: _____ Date: _____

Collaborators: _____ Lab Section: _____

PRELAB EXERCISES (2 points)*This prelab must be completed and handed in to the lab TA at the start of the lab.***Question 1****1 points**

You have just completed the first part of this lab and have five time values for a particular height: 1.8 s, 1.7 s, 1.9 s, 0.8 s, and 1.9 s.

- (a) ($\frac{1}{2}$ point) Give one quantitative reason why 0.8 s is or is not consistent with the other measurements.
- (b) ($\frac{1}{2}$ point) If 0.8 s is an outlier, what is one explanation for what could have gone wrong in that trial? Use details of the experiment in your answer.

Question 2**1 point**

In this lab you will use Atwood's machine to measure the acceleration due to gravity, g . The machine works by hanging two masses on a pulley, with each mass acted upon by gravity. Since the masses are on opposite sides of the pulley, their weights oppose each other and the net acceleration is less than g . To see this, draw all relevant forces in the diagram below. Be sure to indicate the direction of friction in the pulley. Assume $m_1 > m_2$.

Figure 1: A cartoon of Atwood's machine.

